1 Two coils, C and D, are placed with their axes on a straight line, as shown in Fig. 1.1.

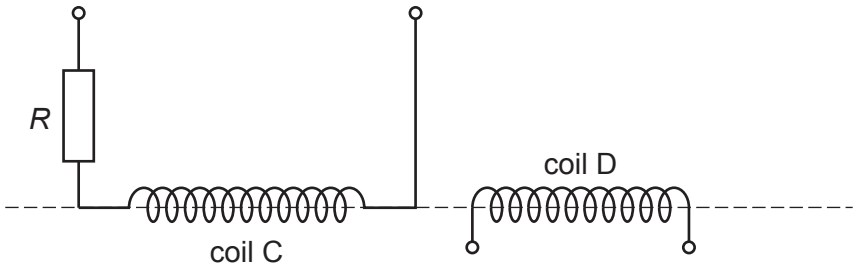

Fig. 1.1

A resistor of resistance R is connected in series with coil C.

A changing magnetic flux of frequency f in coil C causes an electromotive force (e.m.f.) E to be induced across the terminals of coil D.

It is suggested that E is related to f by the relationship

$$E = \frac{pf^qV}{R}$$

where V is the potential difference across the resistor and coil C, and p and q are constants.

Plan a laboratory experiment to test the relationship between E and f.

Draw a diagram showing the arrangement of your equipment.

Explain how the results could be used to determine values for p and q.

In your plan you should include:

- the procedure to be followed
- the measurements to be taken
- the control of variables
- the analysis of the data
- any safety precautions to be taken.

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